Hyperspectral Target Detection: Learning Faithful Background Representations via Orthogonal Subspace-Guided Variational Autoencoder

Qu Tian, Chengxun He, Yang Xu, *Member, IEEE*, Zebin Wu, *Senior Member, IEEE*, and Zhihui Wei, *Member, IEEE*

***Abstract*— Hyperspectral image (HSI) target detection plays a pivotal role in both military and civilian sectors. Nevertheless, this task is fraught with challenges because of the limited availability of target samples and the intricate nature of the background within real-world HSIs. In this study, we present an innovative background learning model based on the orthogonal subspace-guided variational autoencoder (VAE), tailored to discern the background distribution in hyperspectral imagery. Given the scarcity of target samples, our model is exclusively trained on background spectral samples, enabling precise modeling of the background distribution. The crux of our approach lies in detecting disparities between the reconstructed HSI and the original HSI, providing a mechanism for faithful target identification. To procure background samples, a coarse detection of the test HSI is first conducted. However, this process proves challenging, as obtaining pristine background pixels is a formidable task. To mitigate the influence of suspicious target samples on the background reconstruction, we employ orthogonal subspace loss on the reconstructed HSI. Extensive experiments conducted on four real-world HSIs substantiate that the proposed framework performs highly competitively and the results outperform other state-of-the-art HSI target detection methods. The source codes of this study are available at https://github.com/CX-He/OS-VAE.**

***Index Terms*— Background learning, hyperspectral image (HSI), orthogonal subspace, target detection, variational autoencoder (VAE).**

I. Introduction

ABUNDANT spectral information contained in the hyperspectral image (HSI) serves as the cornerstone for the success of many high-level visual tasks, including target detection [1], [2], [3], anomaly detection [4], [5], [6], land cover classification [7], [8], [9], and change detection [10], [11], [12], among others. For example, HSI target detection leverages the discriminative information contained in the given prior spectrum to accurately identify potential targets of interest at the pixel-level or even subpixel-level within complex backgrounds. As a result, the development of HSI target detection methods has garnered considerable attention in diverse practical application scenarios, including remote sensing, military reconnaissance, land and sea target observation, mineral surveying, and criminal investigation [13].

A. Backgrounds

Strong application demands and rapid technological advancements have driven the development of a series of HSI target detection methods. For example, constrained energy minimization (CEM) is a popular one that enforces constraints on the target subspace to generate a finite impulse response filter [14]. The output energy of this filter is minimized, thereby realizing the suppression of the background's response. Many variants of CEM have been proposed to more accurately characterize the intrinsic properties of targets and backgrounds, owing to its advantages of fast speed and high accuracy. Hierarchical CEM is one of the most representative improvement methods [15]. Through a hierarchical structure that better suppresses the background, it significantly enhances the detection rate layer by layer. Another inspiring variant is the ensemble-based CEM [16]. The cascaded detection and multiscale scanning strategies are introduced to improve the detection results. In the diverse direction CEM method [17], an adaptive neighborhood feature aggregation strategy is employed to comprehensively assess the importance of different directional neighborhood information. Orthogonal subspace projection (OSP) [18] is another emblematical detector that can further suppress unimportant spectral features while reducing the dimensionality of HSI. In other words, it begins by projecting each pixel vector into the orthogonal subspace of the background. Then, the residual is projected onto target vectors of interest, leading to an improved signal-to-noise ratio. The spectral angle mapper-based method [19] identifies targets by calculating the spectral difference between target and background pixels. Additionally, several methods

Manuscript received 2 January 2024; revised 25 March 2024; accepted 19 April 2024. Date of publication 26 April 2024; date of current version 7 May 2024. This work was supported in part by the National Natural Science Foundation of China under Grant U23B2006, Grant 92370203, and Grant 62071233; in part by Jiangsu Provincial Innovation Support Program under Grant BZ2023046; in part by Jiangsu Provincial Natural Science Foundation of China under Grant BK20211570 and Grant BK20170858; in part by Jiangsu Provincial Key Research and Development Program under Grant BE2022065-2; and in part by the Open Foundations of Jiangsu Province Engineering Research Center of Airborne Detecting and Intelligent Perceptive Technology under Grant JSECF2023-03. *(Qu Tian and Chengxun He are co-first authors.) (Corresponding author: Yang Xu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3393931

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based on the linear mixture model distinguish targets from the backgrounds through spectral decomposition of HSI [20]. Other approaches, such as matched subspace detector [21], adaptive cosine estimator (ACE) [22], [23], and matched filter [24], also exhibit promising values on the HSI detection task.

However, in real-world scenarios, an HSI often cover large areas of the ground scene, where multiple substances with different spectral characteristics can simultaneously appear. Consequently, the practical application of the above detectors poses several challenges. For example, CEM-based detectors may overlook the impact of target pixels when estimating the background pixels; OSP-based ones struggle to effectively handle datasets with complex backgrounds. Meanwhile, some studies typically make certain assumptions about the spectral distribution of ground objects, which might not fully align with real data. ACE-based detectors, for instance, often assume additional noise in the background but not in the target. As a result, accurately extracting the discriminative characteristics of target and background subspaces becomes challenging.

To delve deeper into the intrinsic properties of HSI data, many studies strive to develop more sophisticated modeling techniques [25], [26], [27]. The representative kernel methods and sparse representation methods are successfully applied to address the HSI detection problem. Kernel-based OSP detector [28] enables the conventional OSP method to effectively leverage the nonlinear characteristics of HSI by mapping the original space to the kernel space. However, the main challenge lies in designing appropriate kernels that accurately capture the linear inseparability of the HSI data [29]. In study, Chen et al. [30] proposed a sparse representation method that incorporates an additional smoothness constraint, allowing for the utilization of the inherent piecewise smoothness observed in HSI. The well-known sparse and collaborative representation method [31] constructs test pixels by creating a sparse representation of the target atoms and a cooperative representation of the background atoms. Their results emphasize the importance of dictionary construction in achieving faithful detection results. To this end, study [32] proposes the integration of the clustering technique to construct a more rational dictionary for the background subspace. The learning of the target subspace is translated into an optimization problem of the low-dimensional sparse coefficient. Furthermore, Ling et al. [33] introduced an iterative regularization sparse learning method that focuses on extracting the abundance of subpixel targets, resulting in improved results of target positioning. In general, the above-mentioned methods, which involve complex modeling and optimization problems, face challenges such as difficulty in solving and low efficiency, despite their potential for improving detection rates. Moreover, these methods often struggle to effectively capture the deeper layer features that are implicitly present in redundant high-dimensional HSI data.

Recently, deep learning has made remarkable advancements in various domains, including pattern recognition and computer vision [34], [35], [36], [37], [38], [39]. Developing the data-driven regime to learn complex deep nonlinear features has been verified highly effective in HSI target detection task [40], [41], [42]. This powerful practice enables the acquisition of more sophisticated feature representations, naturally leading to improved performance in the representation of both targets and backgrounds of HSI. From the perspective of learning patterns, deep learning-based HSI detectors can be categorized into supervised and unsupervised approaches. In the supervised family, Li et al. [43] pioneered the design of the first HSI target detection model based on the convolutional neural network (CNN). The successful application of this framework has served as inspiration for numerous research endeavors in this family. Another example of a representative model is HTD-net [44]. This model employs an enhanced autoencoder to generate target features and subsequently utilizes a linear prediction strategy to distinguish background spectra that exhibit significant differences from the target samples. Meanwhile, employing deep networks to learn spectral features between labeled samples is also a useful strategy for enhancing the discrimination between targets and backgrounds. In other words, advanced network models such as Faster R-CNN [45] and vision transformer [46], [47] have also been verified to be effectively adapted to the HSI detection task. Unsupervised models aim to enhance the network's learning ability of target features under certain constraints, thereby alleviating the difficulty of acquiring large-scale, high-quality labeled data samples in real scenarios [48], [49]. Variational autoencoder (VAE) [50], is a highly scalable and ambassadorial unsupervised network. Xie et al. [49] proposed an enhanced VAE network to further capture the nonlinear features of the HSI data. They utilized nodes in the hidden layer of VAE to better represent spectral information. Li et al. [51] designed a new unsupervised detection model based on hierarchical VAE and linear mixing principles, which enhances the target while improving the ability to suppress the background. In a similar vein, Shi et al. [52] incorporated residual learning to develop a novel macro-micro residual autoencoder model designed specifically for HSI target detection. Furthermore, taking the background component into special consideration is deemed important. For instance, in study [53], researchers adapted adversarial autoencoder and employed adversarial learning strategies to estimate the knowledge of background distributions. This indicates that constructing a more accurate background representation can contribute to improved target discrimination.

B. Motivations

While the above deep learning methods have demonstrated significant improvements in enhancing the accuracy of the HSI target detection task, it is crucial to acknowledge the existence of the following imperative challenges.

1) From the perspective of target analysis, the efficacy of the supervised learning methods often hinges on a large number of trainable parameters, necessitating an abundance of high-quality labeled samples for training. However, obtaining a substantial quantity of labeled spectral samples for specific scenes verifies to be an exceedingly difficult task in real-world scene. If only limited prior spectral information is considered,

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it may lead to insufficient generalization ability of the trained model when dealing with cross-scenario applications. Some recent studies adopt unsupervised generative paradigms to augment target samples. For instance, in literature [44], an autoencoder-based approach is considered. However, we have observed that this approach, lacking appropriate constraints, often leads to generated samples that struggle to robustly represent the widespread subpixels. Specifically, the mixture of target pixels and background pixels. Accordingly, this deficiency becomes more pronounced as the training process advances, ultimately resulting in a learned detector with limited capability to identify targets form subpixels. How to improve the detection performance in such important scenes with scarce target samples is a key issue that requires to be further studied.

2) From the perspective of background characterization, the modeling and learning of background samples also play a crucial role in effectively distinguishing targets from backgrounds. However, in real-world applications, the background subspace in an HSI is often underlabeled, which leads to existing deep learning-based detectors insufficiently considering background information. A recently proposed background learning method, based on a specific target suppression constraint [53], aims to address this issue by training the network using a new loss function. Specifically, the spectral angle distance (SAD) between the prior target spectrum and the samples is utilized as a metric to acquire pure background samples. Nevertheless, such an approach has limitations in its consideration of a single prior target spectrum with limited information. In scenarios where target pixels are heavily mixed with complex background pixels, it may struggle to effectively separate them from each other.

C. Contributions

Motivated by the aforementioned concerns, this paper is dedicated to addressing the following interesting and essential questions: How can we learn a faithful representation for the background subspace in HSI to effectively alleviate the issue of insufficient labeled samples? Can well-established theoretical results be harnessed to enhance the modeling capabilities of deep networks for HSI target detection tasks? The primary results of our work are to provide positive answers to these queries. Specifically, we propose a novel semisupervised learning method tailored to the HSI target detection task. The aim is to enhance the accuracy of background representation and learning, particularly when confronted with a scarcity of high-quality labeled samples. In short, the main contributions of this study are threefold.

1) We propose the incorporation of the essential concept of OSP into a VAE network. This novel integration aims to enhance the model's capacity to effectively learn the accurate distribution from intricate background information. By projecting the test image onto the orthogonal subspace of the background endmembers, we are able to substantially decrease the possibility of target subpixels being present within the background samples. Consequently, this facilitates the training of a more precise and reliable background representation model.
2) To enhance the representation capability of the OSP operator within the network, we devised a novel background endmembers generation approach. It allows us to acquire a diverse set of background training samples, which served as inputs to the model and provided guidance during the learning process of background representations. By incorporating these background training samples with rich information, our model is able to more effectively capture the spectral disparities between the original HSI and its reconstructed counterpart.
3) To validate the efficacy of our proposed method, we conducted comprehensive experiments and evaluations. The observed results unequivocally demonstrate that our model outperforms existing state-of-the-art methods for HSI target detection. Notably, our proposed model exhibits remarkable adaptability to real-world scenarios characterized by complex background information. Moreover, it significantly enhances the detection rate when faced with limited training samples. These findings provide strong evidence supporting the effectiveness and practical applicability of our method in real applications.

D. Organizations

The remainder of the article is organized as follows. Section II presents the preliminaries of the OSP operator and VAE network. Section III provides a detailed description of the proposed novel HSI target detection method. In Section IV, we report and analyze experimental results obtained on four real-world HSI datasets. Finally, the conclusions and future work are summarized in Section V.

II. Notations and Preliminaries

A. Notations

Hereinafter, scalars, vectors, matrices, and tensors are denoted by x, $\mathbf{x}$, $\mathbf{X}$, and $\mathcal{X}$, respectively. For example, the third-order HSI tensor is signified by $\mathcal{X} \in \mathbb{R}^{h\times w\times l}$, where h, w, and l are the height, width, and spectral bands, respectively. The unfolded spectral matrix of $\mathcal{X}$ and the prior target spectrum are signified by $\mathbf{X} \in \mathbb{R}^{l\times n}$ and $\mathbf{d} \in \mathbb{R}^{l\times 1}$, where n is the pixels number of HSI, that is, $n = h \times w$.

B. Orthogonal Subspace Projection

In this section, we briefly introduce OSP. In the approach proposed in Section III, OSP is reconsidered to be integrated into the learning network. According to the principle of spectral linear mixing, a spectral vector with a number of bands l can be expressed as follows:

$$\mathbb{R}^{l\times 1} \ni \mathbf{x} = \mathbf{M}\mathbf{a} + \mathbf{n} \tag{1}$$

where $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_c]$ is a endmember spectral matrix of size $l \times c$, $\mathbf{m}_i$ is the label of the ith endmember spectrum,

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and $\mathbf{a} = [a_1, a_2, \ldots, a_c]$ is an abundance score vector of size $c \times 1$. The ith element a_i of $\mathbf{a}$ represents the abundance score corresponding to $\mathbf{x}_i$. The random modeling error is denoted as $\mathbf{n} \in \mathbb{R}^{l\times 1}$. It is reasonable to further decompose the spectral bases matrix $\mathbf{M}$ in (1) into two components: the target label vector $\mathbf{d} \in \mathbb{R}^{l\times 1}$ and the background label matrix $\mathbf{U}$, i.e., $\mathbf{M} = [\mathbf{d}, \mathbf{U}]$. Assuming, without losing generality, that $\mathbf{d}$ is the initial endmember label $\mathbf{m}_1$ and $\mathbf{U}$ consists of $[\mathbf{m}_2, \ldots, \mathbf{m}_c]$. The abundance scores of the background and target endmembers are denoted by $\mathbf{a}_\mathbf{U}$ and $\mathbf{a}_\mathbf{d}$, respectively, that is, $\mathbf{a} = (\mathbf{a}_\mathbf{d}\mathbf{a}_\mathbf{U})$. Thus, (1) can be written as

$$\mathbf{x} = \mathbf{d}\mathbf{a}_\mathbf{d} + \mathbf{U}\mathbf{a}_\mathbf{U} + \mathbf{n}. \tag{2}$$

The first step in obtaining a target detection operator is to eliminate the effect of the interfering background vectors $\mathbf{U}$. Therefore, the projection matrix can be represented by

$$\mathbf{P}_\mathbf{U}^\perp = \mathbf{I} - \mathbf{U}\left(\mathbf{U}^T\mathbf{U}\right)^{-1}\mathbf{U}^T \tag{3}$$

where $\mathbf{I}$ is the identity matrix and $(\mathbf{U}^T\mathbf{U})^{-1}\mathbf{U}^T$ is the pseudoinverse of $\mathbf{U}$. Under the assumption of multiple undesired background vectors, the OSP detector can be written as

$$\mathbf{P}_{\text{OSP}} = \mathbf{d}^T\mathbf{P}_\mathbf{U}^\perp. \tag{4}$$

A constant term $(\mathbf{d}^T\mathbf{P}_\mathbf{U}^\perp\mathbf{d})^{-1}$ should be added to account for the estimation accuracy when the OSP is considered as an abundance estimator [54], [55]. Accordingly, the OSP detector in (4) can be written as

$$\mathbf{P}_{\text{OSP}} = \mathbf{d}^T\mathbf{P}_\mathbf{U}^\perp\mathbf{d}^{-1}\mathbf{P}_\mathbf{U}^\perp\mathbf{d}. \tag{5}$$

C. Variational Autoencoder

Thanks to its powerful ability to learn latent representations of complex data, VAE has found extensive applications in various computer vision and image processing tasks. As an improvement over the classical autoencoder, VAE not only inherits the excellent characteristics of unsupervised learning but also provides more accurate learning of deep data distribution. A typical VAE comprises two main components: the encoder and the decoder. The encoder takes an input set and processes it through multiple hidden layers to generate a compressed coding distribution, which retains sufficient information for subsequent network components. On the other hand, the decoder reconstructs the distribution of the hidden layers to generate a complete input representation. Both the encoder and decoder are jointly trained using backpropagation, enabling the optimization of network parameters and the generation of effective encodings for specific tasks. It is important to note that since the dimensions of the hidden layer are usually lower than the input, the encoder needs to filter and discard irrelevant information during encoding. Throughout the training process of VAE, the encoder learns to retain as much relevant information as possible within the constrained encoding while intelligently discarding irrelevant network components.

Given an input spectral vector $\mathbf{x} \in \mathbb{R}^{l\times 1}$, VAE maps it to the hidden layer through the encoder, while the learned features $\mathbf{z}$ are mapped to the output layer by the decoder with a nonlinear activation function. Both components are trained to minimize the reconstruction loss $\mathcal{L}_{\text{recon}}$ between the output $\hat{\mathbf{x}}$ and the input $\mathbf{x}$. Specifically, $\mathcal{L}_{\text{recon}}$ can be formulated as

$$\mathcal{L}_{\text{recon}} = \left\|\hat{\mathbf{x}} - \mathbf{x}\right\|_2. \tag{6}$$

To further regularize the hidden space, VAE makes appropriate adjustments to the encoding–decoding process: instead of directly mapping the input code to points in the hidden space structure, VAE converts it into a probability distribution within the hidden space structure. In other words, VAE utilizes an encoder $\mathcal{Q}(\mathbf{z}|\mathbf{x})$ to approximate the posterior distribution of the latent space, and a decoder $\mathcal{P}(\mathbf{z}|\mathbf{x})$ is employed to model the likelihood of the input. The final objective is to approximate $\mathcal{Q}(\mathbf{z}|\mathbf{x})$ with $\mathcal{P}(\mathbf{z}|\mathbf{x})$ by minimizing the Kullback–Leibler (KL) divergence $\mathcal{D}_{\text{KL}}(\bullet)$ between the two distributions. Therefore, the overall loss of the VAE can be derived as

$$\mathcal{L}_{\text{VAE}} = -\mathcal{D}_{\text{KL}}(\mathcal{Q}(\mathbf{z}|\mathbf{x})\|\mathcal{P}(\mathbf{z})) + \mathcal{L}_{\text{recon}}. \tag{7}$$

Typically, it is assumed that the prior distribution follows a normal distribution in VAE, and the decoder samples from these normal distributions to generate latent vectors. However, directly sampling from these distributions poses challenges for backpropagation through random sampling. To address this issue, a technique called reparameterization was further devised. This involves first sampling a latent vector $\mathbf{z}$ from a standard normal distribution and then transforming it using the following equation:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon} \tag{8}$$

where $\boldsymbol{\sigma}$ and $\boldsymbol{\mu}$ denote the standard deviation vector and the mean vector obtained from the encoder, $\boldsymbol{\varepsilon}$ is the random noise vector sampled from a standard normal distribution, and $\odot$ denotes the Hadamard product.

Finally, in order to ensure appropriate training and stable network, it is necessary to impose KL constraints on the mean and standard deviation of the encoder. As a result, the loss function can be expressed using the following equation:

$$\mathcal{L}_{\text{VAE}} = -\frac{1}{2}\sum_{k=1}^{K}\left(\log\left((\sigma_k)^2\right) + 1 - (\sigma_k)^2 - (\mu_k)^2\right) + \mathcal{L}_{\text{recon}}. \tag{9}$$

where K represents the dimensionality of the latent vector $\mathbf{z}$. The unique characteristics of the VAE make it particularly suitable for generative modeling. Their latent space is intentionally designed to be continuous, allowing for random sampling and interpolation. Therefore, through optimization of the loss function, the VAE offers the potential to generate reconstructed background spectra that exhibit strong similarity to the original spectra in HSI target detection applications. This enables the generation of highly similar representations when reconstructing the background from latent variables.

III. Proposed Orthogonal Subspaces-Guided VAE for Target Detection

In this section, we introduce a novel framework for target detection using the semisupervised VAE network with a novel

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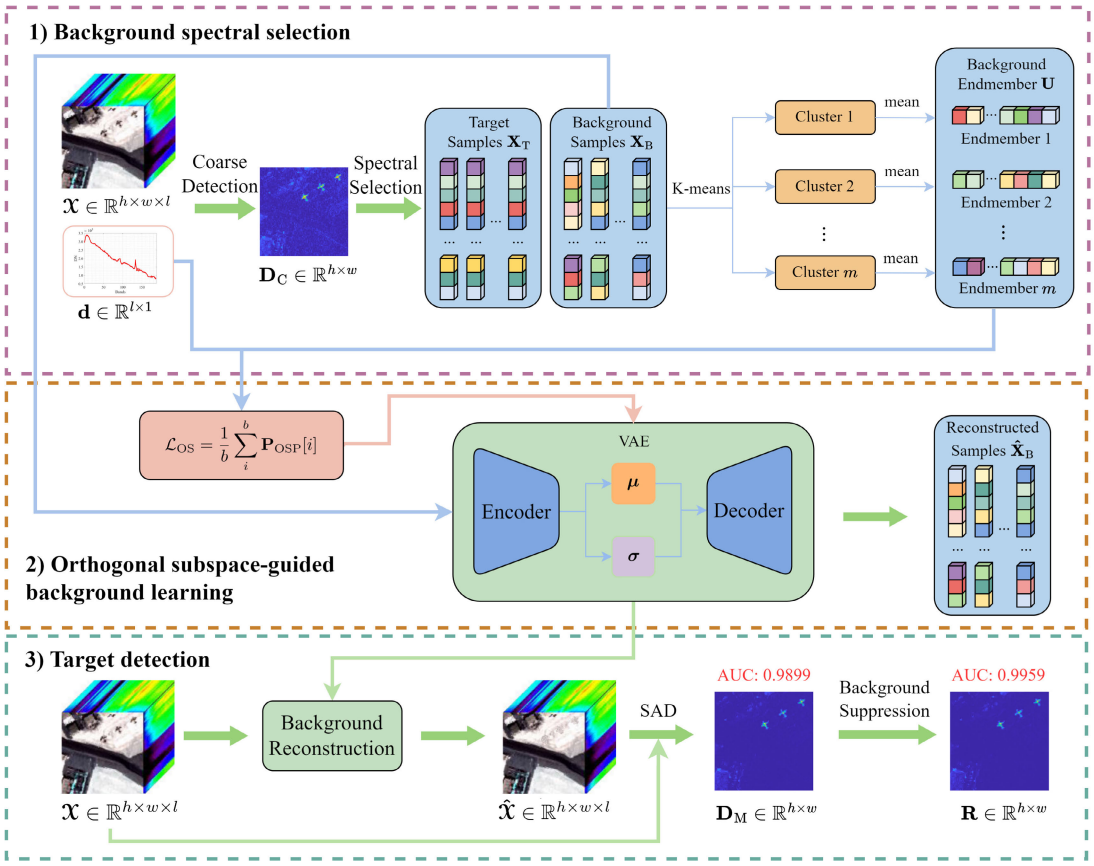


Fig. 1. Schematic of the proposed target detection method.

orthogonal subspace constraint. We integrate the idea of OSP into the VAE to train a more accurate background representation model. Since OSP relies on background endmember information, we have devised a simple yet effective method to generate background endmembers as prior information. Then, background learning is established to address the challenge of limited target samples. Fig. 1 presents the schematic of the proposed OS-VAE framework.

A. Background Spectral Selection

The scarcity of target samples poses a challenge when training deep networks. In contrast, the abundance of background samples in an HSI enables simple network models to effectively reconstruct the background. Thus, acquiring an adequate set of background training samples becomes crucial. To address this issue, we employ the fast CEM algorithm as a preliminary detector to obtain an initial coarse training set. This helps identify potential background samples that can be used for training and allows us to tackle the challenge of limited target samples and lay the foundation for further training and analysis. Given a flattened spectral matrix $\mathbf{X} \in \mathbb{R}^{l\times n}$ and the prior spectrum $\mathbf{d} \in \mathbb{R}^{l\times 1}$, the output response of the CEM filter is given by

$$\mathbf{y} = \left(\hat{\mathbf{w}}\right)^T \mathbf{X} = \frac{\mathbf{d}^T\mathbf{R}^{-1}}{\mathbf{d}\mathbf{R}^{-1}\mathbf{d}^T}\mathbf{X} \tag{10}$$

where $\mathbf{X} = (1/n)\mathbf{X}\mathbf{X}^T$ is the correlation matrix, and $\hat{\mathbf{w}}$ is the obtained optimal finite impulse response filter. The background subspace can be roughly identified by analyzing the normalized detection map $\mathbf{D}_C \in \mathbb{R}^{h\times w}$ reshaped from the obtained response vector $\mathbf{y}$.

Ideally, the response value corresponding to the background pixel in $\mathbf{D}_C$ should be close to 0, indicating its belongingness to the background subspace. Conversely, the response values for target pixels tend to be close to 1, indicating their association with the target subspace. To filter out target pixels as effectively as possible while preserving background samples, we create a filtered $\mathbf{D}_C$ using a lower threshold value ε. Subsequently, as shown in first row of Fig. 1, we select background training samples based on the following formula:

$$\mathcal{X}_B(i,j,:) = \begin{cases} \mathcal{X}(i,j,:), & \mathbf{D}_C(i,j) \le \varepsilon \\ 0, & \mathbf{D}_C(i,j) > \varepsilon \end{cases} \tag{11}$$

where the spectral sample at location (i, j) is represented as $\mathcal{X}(i, j, :)$. That is, the background training samples chosen from $\mathcal{X}$ according to $\mathbf{D}_C$ is denoted as $\mathcal{X}_B$. Through the selection of this coarse detection process, relatively pure background samples that are less affected by interference from the targets can be quickly obtained. Likewise, the remaining samples are selected as target samples. In other words, we used only one prior spectrum $\mathbf{d}$ to initially achieve a reasonable construction of training samples.

However, another crucial aspect that demands attention is the requirement of background endmember information before applying OSP ideas. To address this, we propose a novel strategy for identifying background endmembers from background samples. The overall process of this strategy can be described as follows.

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Step 1: Cluster the background samples into m clusters using the simple yet effective K-means method, considering its completely unsupervised nature.

Step 2: Discard clusters smaller than the spectral dimension to avoid potential inclusion of target pixels or abnormal pixels that could compromise the purity of the background samples.

Step 3: Compute the mean value within each of the remaining clusters. These means represent individual background endmembers. Collectively, they form the background matrix denoted as $\mathbf{U}$.

Remark 1: We emphasize the importance of this strategy in pixel-level HSI target detection that relies on a single spectrum. In general, most pixels in an HSI collected from real scenes are considered as background pixels, whereas the distribution of target pixels is seen as sparse. These sparse target samples often interfere with the representation of background samples. We effectively address this issue through this strategy: by easily removing a limited number of sparse target samples from the clustering results, we can ensure that the resulting background endmembers possess a higher purity.

By following this strategy, we achieve the accurate construction of background endmembers while emphasizing simplicity and efficiency. This minimizes the interference from target elements and provides more accurate information for the following orthogonal subspace-guided background learning.

B. Orthogonal Subspace-Guided Background Learning

We now denote the target set and background set of the spectral matrix $\mathbf{X} \in \mathbb{R}^{l\times n}$ as $\mathbf{T}$ and $\mathbf{B}$, respectively. The ith target spectrum and the jth background spectrum are denoted as $\mathbf{t}_i \in \mathbb{R}^{l\times 1}$ and $\mathbf{b}_j \in \mathbb{R}^{l\times 1}$, respectively. Therefore, we have $p+q=n$, $\mathbf{T}=[\mathbf{t}_1,\mathbf{t}_2,\ldots,\mathbf{t}_p]$, and $\mathbf{B}=[\mathbf{b}_1,\mathbf{b}_2,\ldots,\mathbf{b}_q]$. Similarly, we denote the reconstruction result output by VAE as $\hat{\mathbf{X}} \in \mathbb{R}^{l\times n}$. It also consists of two parts, namely, the reconstructed target $\hat{\mathbf{T}}$ and the reconstructed background $\hat{\mathbf{B}}$. The trained model focuses on learning an exact background representation. Accordingly, as shown in the second row of Fig. 1, the background spectrum is used as input to the VAE, and the output of the encoder is the hidden representation. The encoding process for the target and background subspaces can be formulated as

$$\begin{aligned}\boldsymbol{\mu}_i^t, \boldsymbol{\sigma}_i^t &= \mathcal{F}\left(\mathbf{W}_\mathrm{e}\mathbf{t}_i + \boldsymbol{\beta}_\mathrm{e}\right)\\ \boldsymbol{\mu}_j^b, \boldsymbol{\sigma}_j^b &= \mathcal{F}\left(\mathbf{W}_\mathrm{e}\mathbf{b}_j + \boldsymbol{\beta}_\mathrm{e}\right).\end{aligned} \tag{12}$$

where $\mathbf{W}_\mathrm{e}$ and $\boldsymbol{\beta}_\mathrm{e}$ denote the weight matrix and bias term of the encoding layers, $\boldsymbol{\mu}_i^t$, $\boldsymbol{\sigma}_i^t$, $\boldsymbol{\mu}_j^b$, and $\boldsymbol{\sigma}_j^b$ denote the ith and jth mean and standard deviation vectors obtained from the input target samples and background samples, respectively. $\mathcal{F}$ stands for the LeakyRelu activation function [56]. Then, we can denote the learned feature corresponding to the ith target sample and the jth background sample are denoted as $\mathbf{z}_i^t$ and $\mathbf{z}_j^b$, respectively. They can be obtained from random sampling of the normal distribution $\mathcal{N}(\boldsymbol{\mu}_i^t, \boldsymbol{\sigma}_i^t)$ and $\mathcal{N}(\boldsymbol{\mu}_j^b, \boldsymbol{\sigma}_j^b)$.

With $\mathbf{z}_i^t$ and $\mathbf{z}_j^b$, the reconstructed target and background vectors $\hat{t}_i$ and $\hat{b}_j$ can be obtained by decoding them, that is,

$$\begin{aligned}\hat{t}_i &= \mathcal{F}\left(\mathbf{W}_\mathrm{d}\mathbf{z}_i^t + \boldsymbol{\beta}_\mathrm{d}\right)\\ \hat{b}_j &= \mathcal{F}\left(\mathbf{W}_\mathrm{d}\mathbf{z}_j^b + \boldsymbol{\beta}_\mathrm{d}\right).\end{aligned} \tag{13}$$

where $\mathbf{W}_\mathrm{d}$ and $\boldsymbol{\beta}_\mathrm{d}$ denote the weight matrix and bias term of the decoding layers. Next, we iteratively update the parameters until the VAE network model converges.

As previously mentioned, the primary objective of the model is to learn an accurate background distribution. To accomplish this goal, the proposed model incorporates two essential losses: the orthogonal subspace guided loss $\mathcal{L}_\mathrm{OS}$ and the VAE loss $\mathcal{L}_\mathrm{VAE}$. The former is designed to suppress target signals while reconstructing the background, effectively leveraging the classic OSP idea within the deep learning network. This new loss ensures that the network can effectively differentiate between targets and backgrounds. The latter assesses network performance by calculating the discrepancy between the input spectrum and the output spectrum. As shown in the second row of Fig. 1, through combining the above two loss, the proposed model achieves its design objective of accurately modeling the background distribution.

1) *Orthogonal Subspace-Guided Loss:* The background samples obtained during coarse detection may not be entirely pure, as they may contain undetected target samples. Consequently, to reduce the influence of potential target samples, we consider reinterpreting the well-established statistical theories from new perspectives [57]. That is, we propose the following novel orthogonal subspace-guided loss function to enhance the training of deep networks:

$$\mathcal{L}_\mathrm{OS} = \frac{1}{b}\sum_i^b \mathbf{P}_\mathrm{OSP}[i] \tag{14}$$

where $\mathbf{P}_\mathrm{OSP}[i]$ is the detection result of the OSP operator in (5) for each background sample and b denotes the number of background samples. Given that the projection subspace is the orthogonal subspace of the background endmembers, background pixels projected onto this subspace will yield smaller values, whereas target pixels will result in larger values. Consequently, the network model will likely suppress the reconstruction of the target pixels to minimize loss. Simultaneously, as the projected values of background pixels are inherently small, the reconstruction of the background is essentially less affected. In conclusion, by minimizing this loss, the constructed network can more effectively distinguish between targets and backgrounds.

2) *VAE Loss:* According to (9), we incorporate the VAE loss into the proposed model. This loss aims to minimize the discrepancy between the reconstructed data $\hat{\mathbf{X}}_\mathrm{B}$ and the input data $\mathbf{X}_\mathrm{B}$ as follows:

$$\begin{aligned}\mathcal{L}_\mathrm{VAE} = &\frac{1}{b}\sum_i^b\left(\left\|\hat{\mathbf{x}}_i - \mathbf{x}_i\right\|_2^2\right) - \frac{1}{b}\sum_i^b\\ &\times\left(\frac{1}{2}\sum_{k=1}^h\left(\log\left(\left(\sigma_i^k\right)^2\right)+1-\left(\sigma_i^k\right)^2-\left(\mu_i^k\right)^2\right)\right).\end{aligned} \tag{15}$$

The $\hat{\mathbf{x}}_i$ is the reconstructed spectrum of $\mathbf{x}_i$, h represents the dimensionality of the learned feature $\mathbf{z}$, σ_i^k and μ_i^k denote the kth element of the vector of mean

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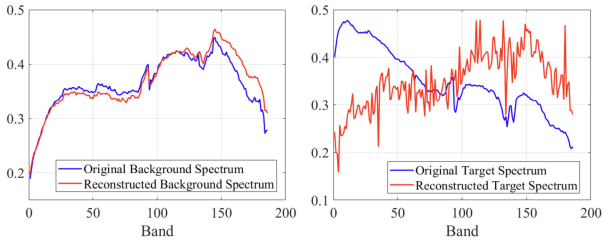


Fig. 2. Comparison of the original and reconstructed spectra samples.

and standard deviation corresponding to the ith learned feature.

Therefore, the overall loss function of the proposed model can be expressed as

$$\mathcal{L} = \rho \mathcal{L}_{\mathrm{OS}} + \mathcal{L}_{\mathrm{VAE}} \tag{16}$$

where a trade-off parameter ρ is used to balance two terms.

Remark 2: Both loss terms in (16) are optimized simultaneously to suppress potential target samples and facilitate more comprehensive learning of background representations. This results in the trained network being less effective in reconstructing target samples since they have not been explicitly trained. Consequently, the network tends to poorly reconstruct target samples while exceling at reconstructing background samples. Such an important and advantageous property significantly enhances our model's capability to differentiate between background pixels and target pixels.

C. Target Detection

As depicted in the third row of Fig. 1, during the target detection process, the spectral matrix $\mathbf{X}$ is fed into the training model to generate the reconstructed $\hat{\mathbf{X}}$. We delve deeper into the efficacy of this reconstruction. Fig. 2 presents the reconstruction results of target and background spectral samples from the San Diego dataset (see Section IV-A for details). The background signature is notably well estimated, while the target signature cannot be reconstructed at all, as expected. As a result, the deviations between the reconstructed $\hat{\mathbf{X}}$ and the original $\mathbf{X}$ can be utilized to identify the presence of a target. Therefore, by evaluating the spectral differences between $\hat{\mathbf{X}}$ and $\mathbf{X}$, targets can be effectively identified. In this study, we employ the SAD function to measure discrepancies between samples pre- and postreconstruction. Using this function brings two distinct advantages: On the one hand, it takes into account both absolute intensity differences and relative shape similarities between spectral signatures, effectively capturing wavelength correlation to facilitate better spectral detail reconstruction. On the other hand, it remains unaffected by overall brightness or radiometric correction of input spectral data, thus enhancing the accuracy and stability of measurement results. For each $\mathbf{x}_i$ and $\hat{\mathbf{x}}_i$, we calculate their distance by

$$d_{\mathrm{M}}^{i} = \cos^{-1}\left(\frac{\mathbf{x}_i^T \hat{\mathbf{x}}_i}{(\|\mathbf{x}_i\|)\left(\|\hat{\mathbf{x}}_i\|\right)}\right) = \cos^{-1}\left(\frac{\mathbf{x}_i^T \hat{\mathbf{x}}_i}{\sqrt{\left(\hat{\mathbf{x}}_i\right)^T \hat{\mathbf{x}}_i}\sqrt{(\mathbf{x}_i)^T \mathbf{x}_i}}\right). \tag{17}$$

Algorithm 1 OS-VAE for HSI Target Detection

Require: HSI tensor $\mathcal{X}$, prior target spectrum $\mathbf{d}$.

1: Compute a coarse detection result $\mathbf{D}_{\mathrm{C}}$ from input $\mathcal{X}$ by (10).
2: Construct the background training set $\mathcal{X}_{\mathrm{B}}$ from $\mathbf{D}_{\mathrm{C}}$ and input $\mathcal{X}$ by (11).
3: **for** each epoch **do**
4: Minimize the loss function $\mathcal{L}$ in (16).
5: **end for**
6: Reconstruct $\hat{\mathcal{X}}$ with the trained model.
7: Compute the spectral distance map $\mathbf{D}_{\mathrm{M}}$ by (17);
8: Compute $\mathbf{R}$ from $\mathbf{D}_{\mathrm{M}}$ and $\mathbf{D}_{\mathrm{C}}$ by (18) and (19).

Ensure: Target map $\mathbf{R}$;

Then, a distance map $\mathbf{D}_{\mathrm{M}} \in \mathbb{R}^{h \times w}$ can be obtained by projecting $\mathbf{d}_{\mathrm{M}} = [d_{\mathrm{M}}^1, d_{\mathrm{M}}^2, \ldots, d_{\mathrm{M}}^{h \times w}]$ into the original dimension.

The overwhelming volume of high-dimensional HSI datasets, coupled with the presence of spectral variability, poses a significant challenge for methods aiming to ensure accurate reconstruction of every sample. In some cases, samples with lower reconstruction accuracy may lead to reduced background suppression in local areas. To address this important issue in real-world applications, we propose a solution that involves merging the coarse detection result with the model's predictive output. Coarse detection is typically effective at suppressing background noise. By employing a nonlinear suppression function, we can suppress pixels with low target response values based on the coarse detection results while retaining those with higher values. The employed nonlinear function is defined as follows:

$$\mathcal{S}(x) = \begin{cases} 1 - e^{-\alpha x}, & x \geq 0 \\ 0, & x < 0 \end{cases} \tag{18}$$

where α is a positive parameter for adjusting the performance of background suppression. For the target detection result, each detection pixel d_{M}^{i} is multiplied by a nonnegative value according to the output of the coarse detection result $\mathbf{y}$ (vectorized $\mathbf{D}_{\mathrm{C}}$) in (10), that is,

$$r_i = \mathcal{S}(y_i) d_{\mathrm{M}}^{i} \tag{19}$$

where r_i is the ith entry of transformed detection result vector $\mathbf{r} = [r_1, r_2, \ldots, r_n]$ and y_i stands for the ith entry of $\mathbf{y}$. One can reshape $\mathbf{r}$ into spatial dimensions of HSI to obtain the final detection result $\mathbf{R} \in \mathbb{R}^{h \times w}$. By employing the nonlinear transformation, we effectively enhance target detection effectiveness while maintaining sufficient background suppression. The complete solution procedures of our target detection model is summarized in Algorithm 1.

IV. Experimental Evaluations

In this section, we perform extensive numerical experiments to verify the effectiveness of the proposed OS-VAE method on HSI target detection task. We proceed by presenting a comprehensive description of the experimental configuration, followed by a comparison of the results against state-of-the-art methods. Additionally, we report a detailed analysis

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of the proposed model, which includes ablation study and hyperparameter analysis.

A. Experimental Configurations

1) Datasets: Four commonly used public datasets for evaluating HSI target detection methods are employed. Their details and included targets are summarized below.

1) *Segundo Dataset [53]:* The captured scene of this HSI encompasses the city of El Segundo, California, USA, with spatial sizes of 250 × 300. It contains 224 spectral bands and covers wavelengths from 366 to 2496 nm. It predominantly includes an oil refining district, a school district, various parks, and residential areas. Notably, the presence of storage tanks and towers, primarily associated with refineries, amounts to 2048 pixels, were considered the desired targets.
2) *San Diego Dataset*[1]*:* This dataset contains 224 spectral bands and covers wavelengths from 370 to 2510 nm. The captured scene encompasses the city of San Diego airport with spatial sizes of 100 × 100. After removing noise bands, 186 bands were utilized in our experiments. Three planes in the image were considered the desired targets.
3) *ABU-Airport dataset [58]:* This dataset contains two different HSIs with spectral channel numbers of 191 and 205, respectively. We denote them as ABU-Airport-I and ABU-Airport-II, respectively. Their spatial sizes are 100 × 100 and several different planes were considered the desired targets.

2) Baselines: We compare with the following popular HSI detectors: CEM[2] based on target maximization and output energy minimization constraints [14], ACE[2] based on adaptive cosine similarity measure [23], CSCR[3] based on collaboration and sparse representation [31], DSC[4] based on clustering dictionary generation and sparse representation [32], BLTSC[5] based on target suppression constraints and background learning [53], and HTD-net[3] based on target sample generation and CNN [44]. All codes are downloaded from the authors' public homepages, and the hyperparameters are adjusted to the optimal according to the corresponding guidance in their articles. Before evaluations, the gray values of all datasets were normalized to the interval [0, 1].

3) Evaluation Criteria: We employ the visualized detection map and the receiver operating characteristic (ROC) curve [59] generated by adjusting a threshold τ to evaluate the detection performance. In 3-D ROC curves, the amounts of target pixels with detection values more than and less than τ are denoted as true positive (TP) and false negative (FN), respectively. In the same way, the amounts of background pixels with detection values more than and less than τ are represented by false positive (FP) and true negative (TN), respectively. The 2-D ROC curves, specifically (P_f, P_d), (τ, P_d), and (τ, P_f), represent the overall detection effectiveness, the target detection effectiveness, and the background suppression effectiveness of different HSI target detection methods, respectively. Specifically, the detection probability P_d and the false alarm probability P_f at each threshold are computed by

$$P_d = \frac{TP}{TP + FN}, \quad P_f = \frac{FP}{FP + TN}. \tag{20}$$

In the case of a method that yields outstanding detection results, the ideal 2-D ROC curves (P_f, P_d), (τ, P_f), and (τ, P_d) should closely approach the upper left corner, lower left corner, and upper right corner of the coordinate axis, respectively. The relationship between the false alarm rate and detection rate can be depicted by the (P_f, P_d) curve. The areas under the 2-D ROC curves, denoted as AUC (P_f, P_d), AUC (τ, P_d), and AUC (τ, P_f), are computed for quantitative evaluation. In the case of AUC (P_f, P_d) and AUC (τ, P_d), higher values signify superior results, where 1 is the optimal value. Conversely, in the case of AUC (τ, P_f), lower value is preferable, where 0 is the optimal value. Furthermore, we consider two combined criteria that incorporate the three AUC values for an overall evaluation. These criteria are denoted as AUC_{OA} and AUC_{SNPR}, and they are expressed as follows:

$$AUC_{OA} = AUC_{(P_f,P_d)} - AUC_{(\tau,P_f)} + AUC_{(\tau,P_d)} \tag{21}$$

$$AUC_{SNPR} = \frac{AUC_{(\tau,P_d)}}{AUC_{(\tau,P_f)}}. \tag{22}$$

Among these two criteria, higher values of AUC_{OA} and AUC_{SNPR} indicate better detection performance. The optimal values for these criteria are 2 and $+\infty$, respectively.

B. Comparative Results on HSI Target Detection

Fig. 3 intuitively displays the detection maps obtained by all competing methods on four real HSI datasets. The results illustrate that our proposed OS-VAE method outperforms other competitors in terms of target identification and background suppression. Classic detection techniques such as CEM and ACE struggle to accurately detect targets while suppressing the background. Likewise, the DSC method exhibits poor performance when processing HSI with complex structural information, resulting in unpleasing false alarm in detection. Regarding background suppression, our proposed method demonstrates superior performance compared to recently proposed methods like HTD-net, CSCR, and DSC. It is worth noting that although ACE excels in background suppression, it tends to misclassify weak targets as unnecessary information. Similarly, the BLTSC method, which also considers background representation, performs well in suppressing the background but slightly lags behind our proposed method. This reaffirms the effectiveness of incorporating the orthogonal subspace idea into the network model. Furthermore, from the results of our OS-VAE method, the difference between target and background (particularly on the San Diego dataset) is significantly larger compared to DSC, HTD-net, and CSCR methods. This indicates that our results closely align with the ground truth. While ACE and BLTSC exhibit better accuracy in target positioning, their results lack complete coverage of

[1]https://aviris.jpl.nasa.gov/data/index.html
[2]https://github.com/safakgunes/Hyperspectral-Images-Target-Detection/tree/main/codes
[3]https://fdss.bit.edu.cn/pub/fsyxhyxtktz/yjdw/js/b153191.htm
[4]https://github.com/shendb2022/DSC
[5]https://github.com/zhangxin-xd/BLTSC

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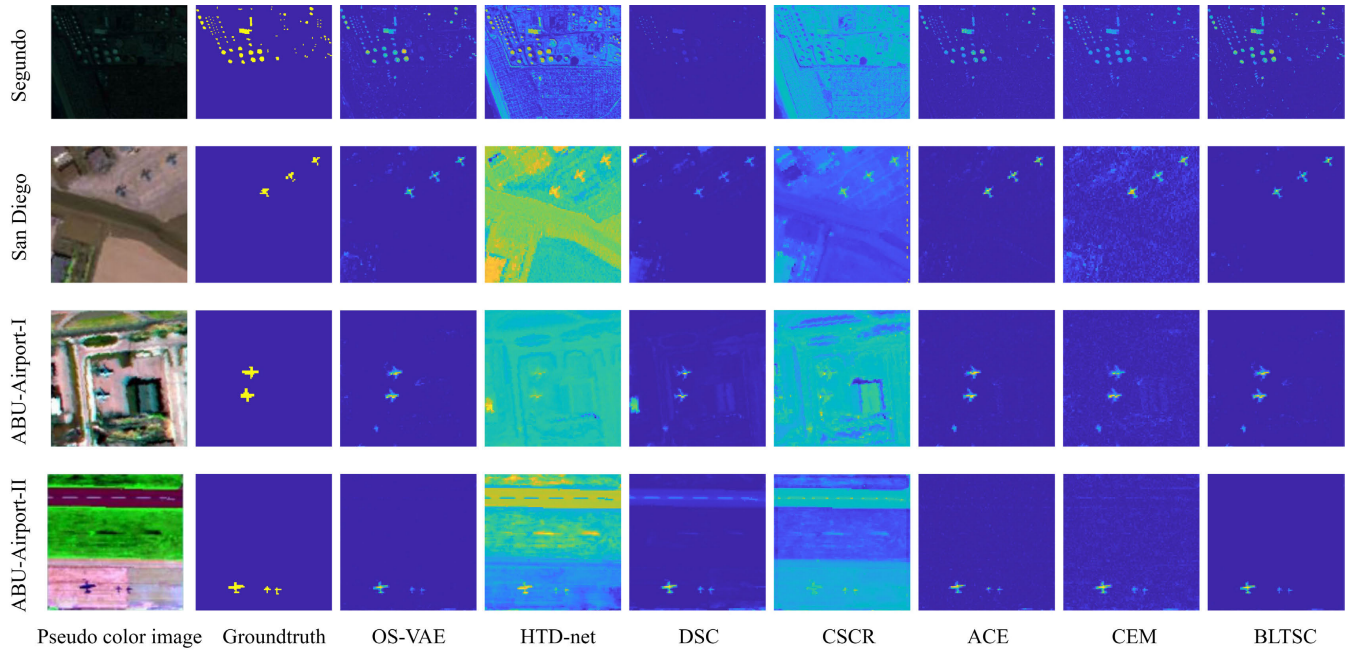


Fig. 3. Pseudo color images of the four datasets, along with their corresponding ground-truth images and the detection results from different methods.

target pixels. Conversely, HTD-net, CSCR, DSC, and CEM produce a significant number of FPs. In summary, our OS-VAE method achieves competitive results in both highlighting targets and suppressing the background. Despite the relatively low purity of background samples in the coarse detection results, which may contain potential target samples, our novel integration of the orthogonal subspace idea into the VAE network effectively removes these potential undesirable samples. As a result, only the background samples are accurately reconstructed, leading to a more precise separation between the target and background subspaces.

We present the ROC curves corresponding to the aforementioned detection maps in Fig. 4. Compared to other methods, the 2-D ROC (P_f, P_d) curve obtained by our OS-VAE method demonstrates a closer position to the upper left corner on the Segundo and ABU-Airport-II datasets. The results on the San Diego and ABU-Airport-I datasets show slightly lower performance. Specifically, when P_f is less than 10^{-2}, the accuracy of this quantitative indicator achieved by OS-VAE is slightly inferior to that of ACE and BLTSC. Nevertheless, upon examining the actual detection maps shown in Fig. 3, it becomes evident that OS-VAE yields more convincing and intuitive detection results across all datasets compared with ACE and BLTSC. Notably, OS-VAE outperforms BLTSC in background suppression and verifies to be more competitive than ACE in target identification. This observation underscores the importance of considering multiple aspects when evaluating detection results. In summary, our proposed OS-VAE method consistently achieves a highly competitive average optimal detection rate across all four datasets.

Although the ROC curves can provide insights into the detection performance, it is often difficult to accurately evaluate the results due to overlapping results from multiple methods. Thus, we present detailed comparisons of the five AUC indicators obtained across the four datasets in Tables I–IV. These results allow for further quantitative analysis of the detection results. Bold font indicates the best results, while underlined ones indicate the second-best results. Upon examination, it becomes evident that our OS-VAE method consistently achieves the highest results for the most crucial $AUC_{(P_f,P_d)}$ score across all four datasets, demonstrating its competitiveness. Even on the San Diego and AUB-airport-I datasets, which exhibit suboptimal ROC curves, our proposed method attains the highest quantitative score. While HTD-net demonstrates the best $AUC_{(\tau,P_d)}$ results across the four datasets, a closer look at Fig. 3 reveals that it performs poorly in terms of background suppression. In contrast, despite not achieving the optimal $AUC_{(\tau,P_d)}$ value, our OS-VAE method excels in average optimal $AUC_{(\tau,P_f)}$ on all four datasets. This strongly supports the effectiveness of the proposed background learning approach in suppressing complex backgrounds in HSI. Furthermore, OS-VAE significantly outperforms other methods in terms of the overall evaluation criterion AUC_{SNPR}. Overall, the comprehensive analysis provided by the AUC scores reaffirms the competitiveness of the proposed OS-VAE method in HSI target detection.

We further discuss the target-background separation performance of the proposed method and the baselines on four datasets. As shown in Fig. 5, it is observed that the proposed OS-VAE method demonstrates excellent separation across all four datasets. That is, the targets and backgrounds are well separated with a significant distance between them and minimal overlap. For instance, on the ABU-Airport-I dataset, our OS-VAE exhibits the largest distance between targets and backgrounds and the smallest degree of overlap compared to other methods. Furthermore, in terms of background suppression, it is evident that OS-VAE also successfully suppresses the background, confirming the effectiveness of our background learning design. BLTSC slightly surpasses our method in terms of background suppression, but it is still inferior to

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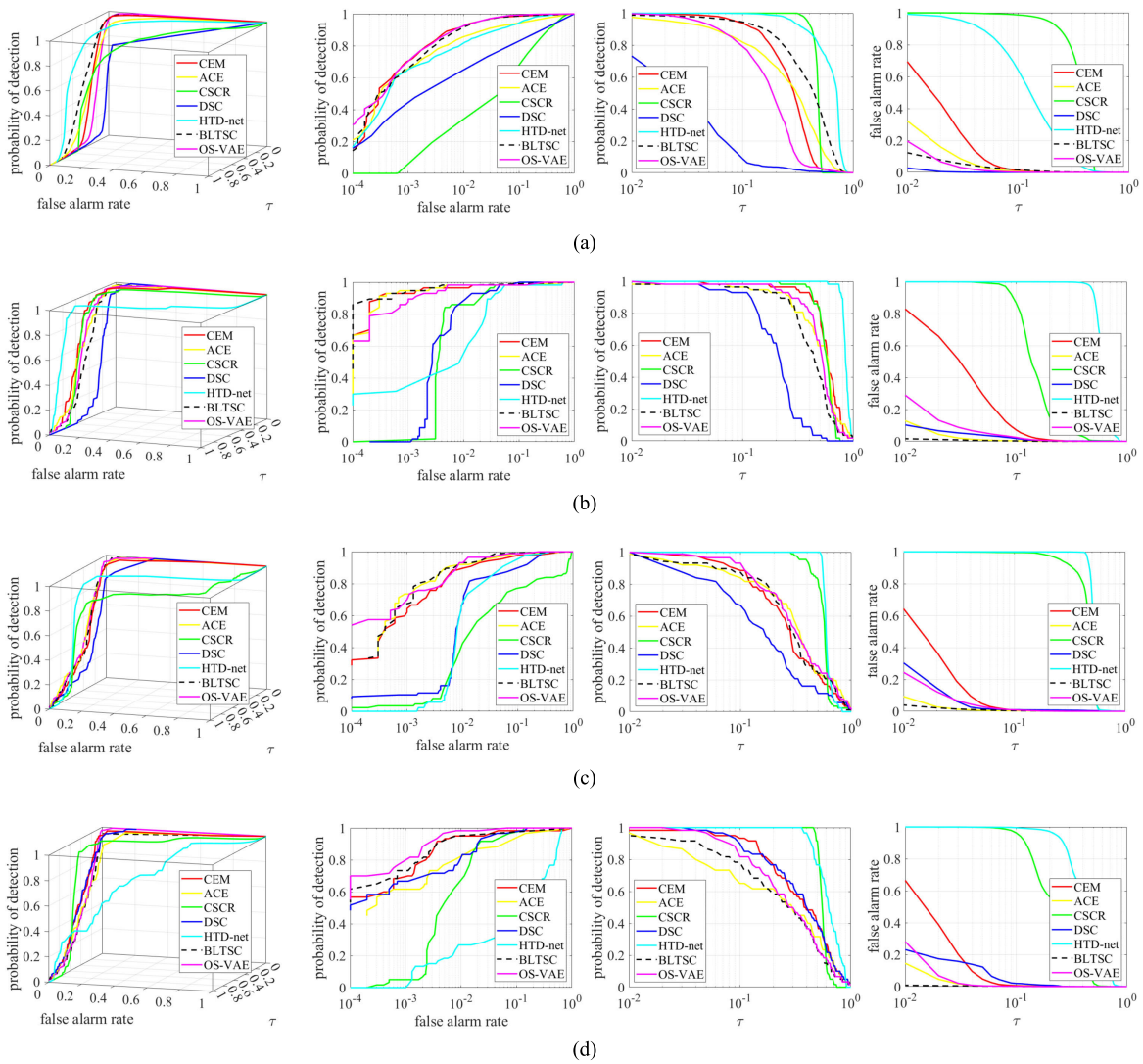


Fig. 4. Three-dimensional receiver ROC curves and the corresponding 2-D ROC curves for various comparative methods on four datasets. From left to right are the 3-D ROC curve, 2-D ROC (P_f, P_d) curve, 2-D ROC(τ, P_d) curve, and 2-D ROC(τ, P_f) curve, respectively. (a) Segundo. (b) San Diego. (c) ABU-Airport-I. (d) ABU-Airport-II.

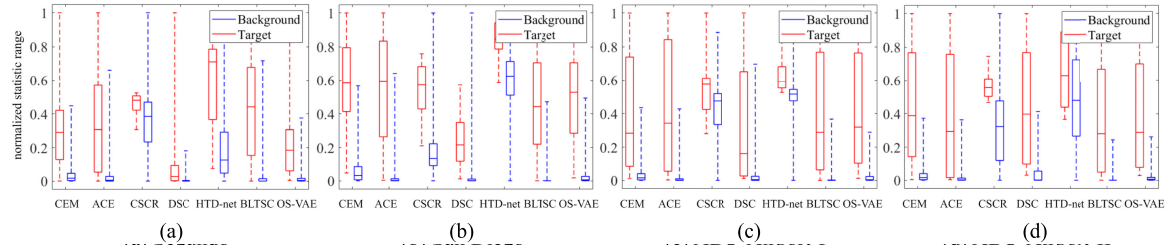


Fig. 5. Target-background separation diagrams of competing methods on four datasets. (a) Segundo. (b) San Diego. (c) ABU-Airport-I. (d) ABU-Airport-II.

our approach in the degree of target-background separation. Overall, the proposed method consistently produces clear, intuitive detection maps, yields excellent average quantitative results, and demonstrates robust target-background separation capabilities across multiple datasets.

C. Ablation Study and Parameter Analysis

1) Effectiveness Evaluation of Our Orthogonal Subspace-Guided Idea: A crucial innovation in the HSI target detection method proposed in this study is the introduction of orthogonal subspace-guidance into the deep network. To evaluate the effectiveness of this novel idea, we conducted ablation experiments and reported the results in Table V. For each dataset, the first row represents the results obtained by incorporating orthogonal subspace guidance, while the second row represents the opposite. The results demonstrate that the quantitative detection results, which consider orthogonal subspace loss, outperform those that do not. This observation validates the powerful ability of the proposed orthogonal subspace-guided loss to promote the VAE network in effectively reconstructing

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TABLE I

QUANTITATIVE DETECTION RESULTS OBTAINED BY ALL METHODS ON **SEGUNDO** DATASET. OPTIMAL AND SUBOPTIMAL ARE MARKED BY BOLD AND UNDERLINED, RESPECTIVELY

Method	$AUC_{(P_f,P_d)}$	$AUC_{(\tau,P_d)}$	$AUC_{(\tau,P_f)}$	AUC_{OA}	AUC_{SNPR}
CEM	0.9916	0.2859	0.0224	1.2551	12.7634
ACE	0.9740	0.3099	0.0108	1.2731	28.6944
CSCR	0.8892	0.4711	0.3658	0.9945	1.2879
DSC	0.9323	0.0488	0.0014	0.9797	34.8571
BLTSC	0.9911	0.4232	0.0070	1.4073	60.4571
HTD-net	0.9821	**0.6352**	0.1528	**1.4645**	4.1571
Proposed	**0.9979**	0.1841	**0.0010**	1.1810	**184.1000**

TABLE II

QUANTITATIVE DETECTION RESULTS OBTAINED BY ALL METHODS ON **SAN DIEGO** DATASET. OPTIMAL AND SUBOPTIMAL ARE MARKED BY BOLD AND UNDERLINED, RESPECTIVELY

Method	$AUC_{(P_f,P_d)}$	$AUC_{(\tau,P_d)}$	$AUC_{(\tau,P_f)}$	AUC_{OA}	AUC_{SNPR}
CEM	0.9939	0.6003	0.0415	**1.5527**	14.4651
ACE	0.9932	0.5402	0.0052	1.5283	105.9216
CSCR	0.9919	0.5689	0.1586	1.4022	3.5870
DSC	0.9929	0.2265	0.0076	1.2118	29.8026
BLTSC	0.9957	0.4588	**0.0010**	1.4535	**458.8000**
HTD-net	0.9775	**0.8551**	0.6166	1.2160	1.3868
Proposed	**0.9959**	0.4218	0.0052	1.4188	82.3269

TABLE III

QUANTITATIVE DETECTION RESULTS OBTAINED BY ALL METHODS ON **ABU-AIRPORT-I** DATASET. OPTIMAL AND SUBOPTIMAL ARE MARKED BY BOLD AND UNDERLINED, RESPECTIVELY

Method	$AUC_{(P_f,P_d)}$	$AUC_{(\tau,P_d)}$	$AUC_{(\tau,P_f)}$	AUC_{OA}	AUC_{SNPR}
CEM	0.9863	0.3446	0.0195	1.3114	17.6718
ACE	0.9920	0.3909	0.0043	**1.3786**	90.907
CSCR	0.8457	0.5589	0.4509	0.9537	1.2395
DSC	0.9678	0.2384	0.0124	1.1938	19.2258
BLTSC	0.9937	0.3649	0.0021	1.3565	173.7619
HTD-net	0.9778	**0.6079**	0.5136	1.0721	1.1836
Proposed	**0.9945**	0.3659	**0.0008**	1.3596	**457.3750**

the background while suppressing the target. In other words, by integrating the OSP idea into network, we maintain a small difference between the reconstructed image's background spectra and the original background spectra, while achieving a large difference between the reconstructed target spectra and the original target spectra. This accurate separation of the target and the background is thus accomplished. Our endeavor not only enhances the effectiveness of existing HSI target detection methods based on deep learning but also highlights a promising direction for future research: a reasonable fusion of existing theoretical foundations and learning paradigms.

TABLE IV

QUANTITATIVE DETECTION RESULTS OBTAINED BY ALL METHODS ON **ABU-AIRPORT-II** DATASET. OPTIMAL AND SUBOPTIMAL ARE MARKED BY BOLD AND UNDERLINED, RESPECTIVELY

Method	$AUC_{(P_f,P_d)}$	$AUC_{(\tau,P_d)}$	$AUC_{(\tau,P_f)}$	AUC_{OA}	AUC_{SNPR}
CEM	0.9839	0.4299	0.0198	1.3940	21.7121
ACE	0.9770	0.3490	0.0052	1.3208	64.1154
CSCR	0.9878	0.5598	0.2905	1.2571	1.9270
DSC	0.9912	0.4308	0.0140	**1.4080**	30.7714
BLTSC	0.9841	0.3345	0.0006	1.3180	557.5000
HTD-net	0.7681	**0.6479**	0.4710	0.9450	1.3756
Proposed	**0.9993**	0.3493	**0.0004**	1.3482	**873.2500**

TABLE V

EFFECTIVENESS ANALYSIS OF THE PROPOSED ORTHOGONAL SUBSPACE-GUIDED IDEA ON FOUR DATASETS. OPTIMAL RESULTS ARE BOLDED

Dataset	$\mathcal{L}_{OS}$	$AUC_{(P_f,P_d)}$	$AUC_{(\tau,P_d)}$	$AUC_{(\tau,P_f)}$	AUC_{OA}	AUC_{SNPR}
Segundo	✓	**0.9979**	**0.1841**	**0.0010**	**1.1810**	**184.1000**
	×	0.9910	0.1474	0.0020	1.1364	73.7000
San Diego	✓	**0.9959**	**0.4218**	0.0052	**1.4188**	82.3269
	×	0.9925	0.4081	**0.0025**	1.3981	**163.24**
ABU-Airport-I	✓	**0.9945**	**0.3659**	**0.0008**	**1.3596**	**457.3750**
	×	0.9887	0.2519	0.0015	1.2391	167.9333
ABU-Airport-II	✓	**0.9993**	**0.3493**	**0.0004**	**1.3482**	**873.2500**
	×	0.9838	0.3159	0.0005	1.2992	631.8000

2) Parameter Analysis: This part discusses the important hyperparameters in the proposed OS-VAE method, including the number of hidden nodes in the VAE network, the depth of the VAE network, regularization parameter ρ in (16), and background suppression parameter in (18).

First, selecting an appropriate number of hidden nodes is crucial for achieving satisfactory detection results. The nodes in the hidden layer preserve key features of the input data, enabling the decoder to leverage these features in generating accurate reconstructed samples. Besides influencing the model's expressive capacity, the number of hidden layer nodes also determines the dimension of the latent space. A proper dimension enhances the network's encoding ability, allowing the learned model to capture the complex structural features. In practical applications, a small number of nodes may lead to information loss within the hidden layer, while an large number of nodes can increase the risk of overfitting. We present quantitative detection results obtained with different numbers of nodes across all datasets in Fig. 6(a). Observation reveals that the proposed method achieves the best quantitative results on all HSIs when the number of nodes is set to 30. Therefore, we adopted a fixed number of 30 nodes for all experiments.

Second, the depth of the network, i.e., the number of encoder and decoder layers in the VAE, also plays a significant role in our method. In general, increasing the network's depth enables better data representation, facilitating the capturing of essential features. However, a network that is too deep may

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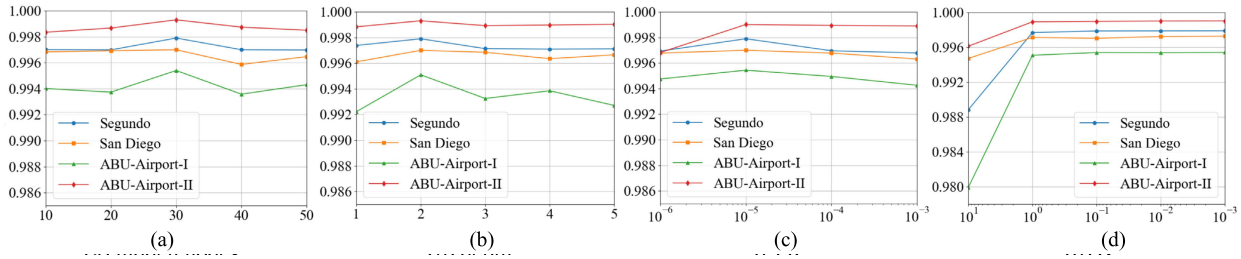


Fig. 6. Sensitivity analysis of different parameters on the detection results of four datasets. (a) Hidden nodes. (b) Depth. (c) ρ. (d) α.

TABLE VI
PARAMETER SETTINGS IN THE PROPOSED METHOD

Parameter	hidden nodes	depth	ρ	α
Value	30	2	$1e-5$	$1e-1$

also encounter issues such as vanishing or exploding gradients. Hence, setting a reasonable network depth strikes a balance between parameter complexity and training efficiency. The results reported in Fig. 6(b) show that a network depth of 2 yields the peak quantitative results across all datasets. Consequently, we fixed a consistent network depth of 2 throughout our experiments.

Third, the regularization parameter ρ governs the trade-off between the VAE loss $\mathcal{L}_{\text{VAE}}$ and the proposed OS loss $\mathcal{L}_{\text{OS}}$. The selection of ρ significantly influences the detection performance. $\mathcal{L}_{\text{OS}}$ is to constrain the reconstruction of targets, ensuring that the background is reconstructed adequately to facilitate a distinct separation between targets and backgrounds. In essence, a larger ρ enhances the network's suppression of target pixels mixed in background pixels. However, setting ρ to an excessively large value may not only impede the reconstruction of targets but also impede the reconstruction of the backgrounds, consequently compromising the differentiation between targets and backgrounds. Hence, assigning a very large value to ρ can undermine the final detection accuracy. Fig. 6(c) indicates that a value of $1e-5$ strikes an optimal balance between two loss term. Hence, we fixed ρ at $1e-5$ for all experiments.

Finally, during the background enhancement stage, an appropriately selected parameter α enables more accurate background suppression, thereby further improving detection accuracy. It represents the degree of background suppression by the nonlinear suppression function. As shown in Fig. 6(d), as α decreases, the effect of background suppression gradually increases. When α decreases to a certain range, the improvement in detection accuracy slows down. If α is too small, the suppression on the background is inadequate, leading to a decrease in detection performance. Therefore, a larger value should be considered. However, although a larger α enhances background suppression, if it is excessively large, the nonlinear suppression function may become ineffective, resulting in the inability to retain potential targets with high values. Based on observations, we set α to $1e-1$, a value that effectively suppresses the background and preserves the target on the four datasets.

We present the overall parameter settings in Table VI. These settings serve as a reference when applying the method in real scenarios. It is essential to select optimal parameters based on specific data characteristics. We recommend considering the parameter settings provided in the table to guide the parameter selection for different datasets.

V. CONCLUSION AND FUTURE WORK

In this study, we focus on the important HSI target detection problem. Theoretically, we present an in-depth analysis of key issues in deep learning models for HSI target detection and further investigate the potential of reinterpreting well-established statistical theory from new perspectives. Technically, we propose a background reconstruction method for HSI target detection that incorporates a VAE network guided by orthogonal subspace concepts. Our proposed method takes into account the challenge of scarce target samples commonly encountered in typical HSI target detection scenarios. We also introduce a novel orthogonal subspace-guided loss to reduce the interference of suspicious target spectra on background spectra reconstruction. This loss is jointly optimized with the VAE loss to facilitate the learning of a more precise representation of the background in HSI. Through extensive experiments conducted on multiple public datasets, we demonstrate the effectiveness of our proposed method in solving the HSI target detection problem. The results validate that our efforts achieve higher detection accuracy and superior background suppression capability compared with state-of-the-art methods.

In future work, we aim to incorporate richer prior information to guide the deep network's ability to express HSI. For instance, we plan to consider incorporating high-order low-rank and sparse properties as additional guidance for the deep network [60], [61]. Additionally, we are interested in designing enhanced deep network architectures specifically tailored for HSI background reconstruction challenges.

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